

ActInf GuestStream 042.1 ~ Bogotá & Djebbara: "Time-consciousness in computational phenomenology"

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hello and welcome it's active inference guest stream number 42.1 on May 10 2023 we're here with Juan Diego Bogota and Zach Jabara we're going to have a presentation time Consciousness in computational phenomenology a temporal analysis of active inference by the authors followed by a discussion so thank you both very much for being here we'll look forward to your presentation and off to you cool thank you so I should start out right brilliant so hello everybody my name is I'm one of the co-authors of this paper I'm a PhD student at the University of Exeter in the UK and kind of my broad interstar philosophical phenomenology and how it can be related to empirical sciences and more specifically to cognitive science and to our understanding of of Consciousness so what we're going to do now is I'm going to give a broad context or paper I'm going to talk about the theoretical and philosophical context that motivated our paper and then zag will talk about kind of the specifics of our analysis of active imprints so let me start now so to kind of situate ourselves in the broader literature uh I guess there's a big big cluster of discussions and topics and approaches within cognitive science and conscious and studies that as it as its names um so that's just trying to explain Consciousness uh scientifically within that kind of big category of approaches there's a very specific set which is um which is trying to connect philosophical phenomenology with the Natural Sciences so we may call this the Naturalization of phenomenology I'll talk about all these in a bit I just want to kind of situate ourselves within the broader context one of these approaches is quite popular in some circles which is called Neo phenomenology and was first introduced or proposed by Francisco arella in the late 90s but recently varela's project has been kind of elaborated on by people within the fep and active nurse community and and they're talking about what it's called computational neural phenomenology or computational phenomenology which as we'll see tries to connect the original project of neurofenaminology of bringing together cognitive science and management but it gives an active inference twist to this whole

story and this is where our project is situated uh our main purpose in the paper is to was to give a temporal analysis of the structure of active inference and in a bit I'll explain you why we thought this was important so uh as you probably know uh Consciousness is a bit of a difficult phenomenon to explain explain scientifically uh we have nowadays a lot of models and theories scientific theories to try to address Consciousness but there's always this kind of lingering worry that there is a gap an explanatory gap between our empirical understanding of the old nature and Consciousness because how we usually do science is try to address and capture empirical phenomena for a third person perspective whereas Consciousness is something that is allegedly given only for a first person perspective so we're on always experiencing the world from our first person perspective and allegedly we don't really have access to to Consciousness as these empirical phenomena we can see in the world so that's kind of an epistemological gap between How We Do Science and how we understand Consciousness without doing much science there's also the so hard problem of Consciousness which is to say that we don't really know how something that Consciousness may arise from say neobiological phenomena or empirical phenomena more generally so this is kind of a general context which shows that Consciousness is quite difficult to understand I don't think we would want to say that it's impossible to you to explain it from a scientific perspective but it's it's somewhat difficult in the broader context of conscious and studies uh since the early 90s there's been some proposals within cognitive scientists of cognitive science and philosophy of mind is trying to connect this whole idea of explaining Consciousness scientifically with what it's known within philosophy as phenomenology so importantly here when we talk about phenomenology we're using a very specific and technical term so in the literature you may distinguish two different ways of talking about phenomenology there's the usual broader sense in which we may mean kind of the what it's like to experience something or the qualitative contents of of experience this is not what we're talking about here rather we're referring to phenomenology which is very specific philosophical tradition which was first proposed by Edward Jose in the early 20th century and was poorly developed by people like Marcus melopenti Martin Heidegger Simon De vavois and there's a long long tradition going on there and what's interesting about phenomenology is that it's simply a philosophical method and way of studying the invariant structures of Consciousness so what all these phenomenologists are trying to say is to capture what they call the essential uh kind of characteristics of what it's like to be conscious of the our lived experience through which we experience the world we do this by describing and analyzing precisely these experiences and

you may think that this is pretty much what uh kind of the broader sense of phenomenal is trying to to get to through this idea of like what the what it's like to experience something but actually phenomenologists are not really interested in the word of experience they're not really interested in the context of of Consciousness so when abnormal is talking about say perceptual experience they're not really interested in the redness of the red or you know the flavor of an apple this kind of what what philosophers or qualia rather they're actually interesting how is it possible that these experiences come to be so this is how it's this refers to these structures when we talk about for example the the temporal structure of Consciousness when we talk about activity embodiment but the important thing here is that we're focusing on the how we experience not what we experience and that's very important because at least how I see it I don't know about Zac uh this town actually constrains the what so it's very different to imagine uh an apple than to experience it so there's a difference between the how we're experiencing the world and and that constrains uh the specific contents of our experience and this is important for the broader context of explaining conscious scientifically because arguably if we're focusing on how we're trying to to capture the essentials of Consciousness and this is a project that was one of the kind of this project was spearheaded by Francisco arellia in the in the 90s but we can actually Trace back this idea to the works of merlot on T and Adam Griffith in the philosophical tradition we can also talk about um Uber drivers and more recently there's a kind of a very broad uh discussion about whether and how can we connect these philosophical considerations with scientific perspectives of consciousness and this brings me to the so-called project of the Naturalization of phenomenal we do science into your world through the land of the natural course we have people understanding of the world instead of metaphysical and epistemological commitments such as that there's a material world out there which isn't get to know through some addicts or doing empirical tests that's that's how we finds very broadly speaking but when you go and look the literature the phenomenological literature what phenomenologists provide are not causal explanations of Consciousness they are simply describing a set of what they call motivational connections which not really um identify with causality which is the the main thing that science is trying to capture causal relationships also there's a issue that phenomenality is not positing empirical entities so when we talk about say unconsciousness we're not talking about empirical thing and empirical object arguably we're talking about a set of processes which and this is kind of the shift from from the scientific issues to those possible ones we're talking about something that is arguably prior to to objectivity so phenomenologists are often reminding the readers that before we're

doing science like we're doing theoretical science we're experiencing the world and we're therefore because it's something that is prior epistemologically to science and to try to capture uh Consciousness scientifically almost to uh commit a category mistake um so given these worries the question is whether we can actually link phenomenology with empirical science such as cognitive science and this question is precisely what we talk about what we call the so-called project of naturalization of phenomenology which is to mean uh how are we to mate to make phenomenology continuous with empirical science there's a lot of proposals in the literature there's a very important book published in 1999 edited by Jean Petitot called naturalizing phenomenology which the editors and several authors discuss how and whether we can actually naturalize phenomenology but one of the most famous proportions is precisely uh Francisco Varela's proposal of the Naturalization of phenomenology which is called neural phenomenology so put very simply what neuro phenomenology is all about is to provide a non-reductive approach to the study of Consciousness this non-reductive approach takes seriously both the analysis and descriptions made by phenomenologists as well as the scientific understanding we have of the neurobiology that um underpins our lived experience these two kind of domains are put in relation to one another by means of what parallel called reciprocal constraints so the idea is that a phenomenology can constrain our understanding of the neurobiological underpinnings and these neural biological underpinnings can also constrain our understanding of experience or phenomenological understanding of experience interestingly and this is where kind of the computational and mathematical aspect comes in Varela also suggested that in order to achieve this kind of circulation between phenomenology and empirical science we have to have some sort of bridge or what it calls a generative passage that allows us to go from these very abstract phenomenological descriptions of experience to the more concrete somatic and the concrete descriptions of somatic and neural processes that underpin lived experience and this very abstract level that allows for this uh passage for this bridge and for him for Varela was the use of dynamical systems theory and more specifically models of non-linear Dynamics so if you look at his 99 paper on time Consciousness what he does is to connect Edmond Husserl's analysis of temporality with certain neural processes by introducing dynamical models into the picture this is what phenomena neuro phenomenology is all about but more recently and uh inspired by this whole project of new phenomenology Maxwell Randstad and colleagues have proposed to further elaborate on this project by employing generative modeling techniques such as active inputs to provide this passage between phenomenology and and the

neurobiology and what is interesting about it is that if you do so introduce genetic modeling techniques what you get is the idea that these have experienced how experienced the world or what phenomenal is called Constitution can be modeled or described in terms of patient inference or approximate basic Librarians an important bit here is that there's a different different ways of interpreting this one way would be to say that Constitution or how experienced the world's literally a matter of basic inference or there's a more kind of instrumentalist or non-realistic conception of this product which would say we can actually just model or describe our lived experience and how experience World in terms of approximated patient inference in this context what happens is that what the the Hat the what what experience in the world is constituted through a set of inferences which are informed by likelihood mappings and you know all these materials that come in place into generative modeling so this kind of a broader context uh that informs our proposal but you may ask why we were focusing on time Consciousness and I think Zach is one's going to say something about this but for now what I just want to say that there's good reasons both phenomenological reasons and scientific reasons to to apply uh these kind of models to time conscious on the one hand and importantly time Consciousness is simply what we mean by time Consciousness our experience of time so how we subjectively experience the passage of time and this this phenomenon Consciousness enjoys a sort of privileged position within uh philosophical phenomenological literature for example who's real heidiagram and leban tea are some prominent phenomenologists who take time conscious to be the most basic structure of Consciousness the point is that all that we're experiencing all the time is always experiencing experienced in time there's always this kind of temporal flow to the way we experience the world and this is kind of uh essential to the way we we engage with the world so if this is the most basic structure of Consciousness and we're trying to understand Consciousness scientifically by modeling through say genetic models or or dynamical System Theory models then probably uh we should start by the most basic level time Consciousness and this is something that several authors have actually tried to do so there's for example both bang elders and barelas dynamical models of time Consciousness which borrow from Edmund husse uh phenomenological analysis there's also gracious computational model of the same processes more recently there's varnia basis a predictive processing model of time Consciousness so if you see there's also a relatively long tradition of trying to capture the Dynamics of time Consciousness in terms of mathematical modeling so now moving on to to our table analysis our main question is whether actually active inference is capable of modeling these Dynamics

in a way that is um that is that makes Justice to the original phenomenological analysis made out by Post Road and that's it for me now I'll give the the floor to Zach here I am and I hope you can hear me and see me yep thanks I'm gonna share my screen all right so I'm gonna pick up here from where uh left um I basically like you said we basically we started talking about well there is this problem um with the flow of experience and um this discreteness in which we we model um well whatever phenomena really especially in cognitive science um so there is this relationship and this claim that we are in fact modeling features of Human Experience like perception memory and so on but it's very few if any of them really take seriously the flow of of Human Experience um so phenomenology is is for philosophers uh and not for scientists apparently so computational phenomenology like JD just talked about is like this new subfield um that takes both seriously the computational approaches but also phenomenology and it's trying to sort of find this middle ground in which we can have a conversation and of course big kudos to to Maxwell for this fantastic Milestone I should I should say before I start that I'm very happy that JD's Juan Diego is here because I think that there is a lot of thoughts and and he has this very cautious way of thinking and basically formulating his sentences that I do not uh have um so it's good that he's here and I should I should probably also say explicitly that I'm I will not be speaking for his behalf at all but we agree on basically all of this so just to be short all right so before I uh dive into the whole temporality and active inference aspect of of um of our paper I would like I would just like to start with a small exotic experiment and I usually give this to to those who don't understand why I personally think in time all the time so uh let's imagine that we have to understand our own experience of of space b this is the space b this is the space a it's a space c so these are all very very different spaces right and imagine there's two like a parallel universe in the first Universe in the first scenario you approach space b from space a right so this is you and you're basically moving in this direction and you're experiencing this space now in Another Universe right could be the fictional you're in this person you're this person and you're approaching space b now I usually ask my students do you think that this experience of space b will be the same and usually they all go like well uh by far the majority says no um it's it's two very very different experiences and they have nothing to do with each other and this is basically what I am what I'm what I'm sort of asking people to sort of start thinking about there is no one-to-one isometric relationship between space and experience and I think this is a very very important and very I wouldn't say obvious but to me it's very clear that they cannot be like that it cannot be that one space or one specific type of content will make me experience the content

the same way every time and I think that the reason for that is change right time Dynamics Everything Changes this is sort of uh what brings me into this will bring us into the discussion about substance philosophy and process philosophy I will not unfold that at all but I will mention a few from time to time a few things about them so here I'm just going trying to make the point that even like intuitively which is usually what we go what we like but we reason with these thought experiments that's basically why they're here and it just appears to us that space is not necessarily uh directly mapped to an because we have to consider time right so I'm asking three questions here and the first one will be how does the continuous flow in Human Experience emerge and I will try to answer that through the differences and similarities between what we call objective and subjective time I will elaborate on that and then I will try to answer how we can we extract principles from personally and temporality so Edmond Russell that that uh Juan Diego just talked about and to answer that I will talk a bit about the enduring present and the integration of two types of temporalities that's the objective and subjective one and then finally what structural answer does active inference comply with these principles sort of you know using phenomenology to come up with a principle constraints and see whether active inference is sort of playing by those rules or not and for that I will try to give you our temporal analysis of active inference as a partially observable Markov decision process a computational model so a pump DP so first things first objective and subject this time so the objective time what do we mean by this and I I like to think of them um as as roughly um what corresponds to substance and process philosophy respectively right because object uh objective time sort of it deals with uh spatial features and everything that is rooted in the space whereas subjective time I like to think has more to do with how things change um so to give an example and a substance philosopher would say that the sun is simply the sun it's a thing and it's a sun whereas a process philosopher would say that no it's a burning Planet it's doing something and so the river is not just a river it's running water and you are not you you're always changing and so on it's heraklitis this is basically the idea and the difference between substance and process velocity now if we project this idea like this abstract idea onto onto the discussion on the relationship and between objective and subject to time um we might we we first of all it's an important discussion because computational models operate on computers and these follow the laws of physics right so we are sort of Bound By by the laws of physics and we sort of have to talk about objective time in in the context of the laws of physics so um I like to think that in space we can have several units at the same time right like you can you can count different kinds of sheep like they don't even have to look

alike or anything like I know that the farmers can tell the difference between the animals but maybe to someone who cannot it will just be like a sheep one sheep two sheep three and so on and you can then even say a number like you can go let's take a sheep number five and you will not even imply any other ship from from well one to four you don't imply them at all you just mean this one right there's an identity versus a similarity uh discussion that needs to be taken up there so space allows us to sort of categorize and discretize the world and I like to think about this as the way that geometric Sim sort of functions where we can take space and just divide it infinitely right you can always take something like anything and you can split it in two then in two again then in two again and you essentially you essentially get what they call the geometric series which is um interestingly how you approach continuous numbers and continuous functions anyway uh of course I mean this uh in in theoretical terms because at some point we're going to reach the plank length length and then we're gonna have to deal with a different kind of problem but in any case my point here is that space allows us in the first place to think in degrees right like if there's a change in degree how much something is but time I think operates in a different way Sam sort of has this single unit right um it's just a single kind as opposed to the degrees so there's a difference in kind when we change moments not a change in degrees which will have in space so there cannot be any division of time this is why I usually say that time for me is completely indivisible it doesn't make sense to to compare two different moments because they have two different units and we cannot compare them so when I say that we measure time I'm basically saying that whenever you measure any time you measure it the same way so when I say five seconds I imply all the seconds between zero and five as opposed to to the space with the Sheep so I guess you can also think about the units of time I'm talking about the measure of time I think that sort of makes it easier to understand but because these computational models they're basically operating with computers that are operating uh with electricity and basically the laws of physics um I think it's important here to to sort of think about that in a broader term um and I wanted to briefly quote uh Sir Arthur Eddington who sort of is famous for introducing the idea of the arrow of time stating that so far as physics is concerned times arrow is a property of entropy alone but I think that there's some truth to that um so if we consider the whole universe as a closed system right then we would expect the Heat and the energy to at some point even out completely thermodynamically speaking right is the is the thermodynamic equilibrium and this is where the entropy is is highest um so because the entropy of the system is maximized or um basically it's it's at its highest value it has reached a state where all the possible ways of Distributing

energy uh between the the particles has been completely exhausted like there's no reason for any particle to move anymore there is no harm cold everything is the same temperature it's the maximum of of uh entropy so um well I guess this is also why you you sort of say that energy is unavailable to perform any work right so um probably I should also mention here that entropy actually is a Greek word for transformation and change and I think this is exactly why uh it fits very well with my understanding of time here uh is to change right it's the process so entropy in a in a abstract way I think it sort of tries to explain not only uh spatial States but also visual stage in relation to time like how they can how they can uh change so uh thermodynamically speaking there will always be a transfer from hot to cold in order to reach maximum entropy to sort of create a balance and uh importantly there is a linear causal relationship right there's always like one damn thing after the other as the saying goes uh there's never anything other than like the linear way and it goes like the past causes the future and so on and of course this is no consensus about this lots of people in quantum physics and quantum mechanics will completely oppose that and think that it actually goes the other way around but typically when we talk about this um at least I believe in Newtonian physics there is a linear cause of relationship so the features of objective time is that it's spatial and that it's temporarily linear now let's move on to the subjective time and as Juan Diego already said it's the ABCs of of uh well it's the a in the ABCs of Consciousness and subjectivity so uh it's going to give a a brief demonstration of how this particular diagram which was developed originally by idmod husal in his uh basically we have to think about the black line here as the objective time sort of right uh one event second event third and fourth and so on and basically if we just take the present here as being the the event B well there is some kind of expectation of what's going to happen at the next moment that is affecting this movement here but not only that there is also another layer to that is going sort of a what we call a pro tension right a protection of what's going to happen even Beyond C and you can actually draw this huge diagram out there and you will eventually get like a kind of protection about the future but at the same time we also have some information about retention right there's something that happened already that we're sort of uh building our present moment based on our retention so the retention sort of explains the immediate past and we do not refer to this as memory or working memory it's very explicit that it's not the same thing um I guess the one way we can think about this is also in in terms of memories uh in terms of Melodies is what I'm trying to say uh and sentences so for example my sentence right now I started it a few moments ago but it's still going and you still have the full sentence in your head and so on and so on um and

basically when we talk about the structure of of human experience and I think that uh Juan Diego's absolutely right when he says that for any naturalization naturalization project for it to succeed it must at the very least get the temple or the right I think this is like the basics of the basics so in this case the way that Edmonton thinks about it is that he he thinks about it as the the living present being made of of a synthesis of retention and protection and this is what we call the Primal impression it's the present moment that will not refer to it always as the Primal impression but that is I believe what do you what you call in in philosophy the technical term for for living present but the the main issue here Still Remains right so how can it's what I'm writing here how can a sequential structure of time be compatible with one that appears to flow like this is this is the the the the the the the the the the issue that we are dealing with here um and I think that there is uh two very important principles that we can point out from here resilient synthesis that any experienced Moment In Time cannot consist of a discrete now moment alone but must instead coexist with the immediate past and immediate future which is basically the retention and the pro tension in this case but there's also a very bergsonian aspect to this which is the interpenetration right so the present moment requires the coming together of subjective and objective temporal structures which is exactly also what you see here there's an objective direction of of uh of time but it's constituted by well past and future all the time and this is basically um if you look at it in in diagrammatic uh or graphical way well the objective time that we call the spatial and linear cause relationship we said that one event causes the next one causes the next one causes the next one and so on the the arrow is very very clear but in subjective time we sort of have this um let's say opportunity to well flip one of the Arabs and say that well if I'm looking at the event B not a and C but just the event B well the event B is constituted by a retention and a protection so we have sort of flipped the arrow here and this sort of um these are two very different orders of time as we call them in our paper and one is characterized by its unidirectional development in time which is the objective one the other one is characterized by bi-directional development in time and this is the subjective one basically what we do um oh probably I should mention here that the notation here is very important to to sort of get that we're always using the superscript for the objective and the subscript for the the for the subject I meant objective obviously the super superscript for objective and subscript but subjective um time dimensions and usually we use T for the superscript and Tau for the subscript just to be sure so what we propose in our paper is basically this integrated continuity where we try to combine both of them into Australian interpretation and obviously we're also relying here a lot on on on um uh conditional

dependencies here and this is from statistics um I guess this is also why I I personally like to think of this as a kind of a tribute to Boltzmann's statistical approach to entropy right because he was the one who sort of took that step away from from uh uh simply thermodynamics and movement in between energy to thinking about it more as uh statistics and probability uh distributions and I think that was very fascinating I think in many ways this is kind of a tribute to that right so taking way of thinking of it and sort of adding this conditional dependency layer to the development of time so um right this is this is what we've sort of uh get at right so if you look at the present moment B this is the event that we're trying to understand well there's a B minus one uh in in in in the objective time but it's actually a pro tension so it's actually a plus one as well because it was a plus one retention later but for the next event which is C and you can actually generalize this uh the way we do at Each corner here so why uh what the sub uh the superscript the object of time plus minus K and the subject of time plus minus 10 and you can do the same for for the other way um the other direction basically what we've been talking about so far is is um talking about systems and whether they're how they function and especially I talked about the universe functioning as kind of a closed system I think that's a huge assumption and I'm not really buying into all of this but it's basically a good way to to explain uh the approaches that we have made but the body open it functions as an open system because it's exchanging energy and information all the time and I'm preaching to the choir here you all know about Market blankets and I don't think that it's important to sort of introduce it to this forum and but I would like to just pinpoint a few things the first one is that I want to remind you that active inference assumes that there's a border between the central nervous system and and the environment and that can be identified through a markup blanket and I also would like to emphasize the fact that it's describing the processes and Dynamics rather than the things themselves and I'm really emphasizing the process versus The Thing and oh okay I'm all right so basically Market blankets they represent the statistical boundaries right um based on on these conditional conditional relationships and they're typically applied in the spatial domain this is my point uh it's it's a statistical boundary in the spatial domain um also I should probably mention that when I talk about conditional dependencies because I realized that there has been uh quite a lot of debate about what we mean when we talk about from the conditional dependencies uh well not specifically for our paper but in general in the statistical work um what we mean here is that there is a relationship between two or more events that that are dependent when a third event occurs this is basically the most generic uh definition of of the conditional dependencies which you can find in in a

Judea Pearl's work on Markov and Bayesian Networks so my point here was just to say that for the Markov blanket you see that um it's just a statistical boundary and I know that Carl's monograph the famous monograph sort of takes that one step further and suggests that well there are Markov blankets actually make up the physical and the spatial boundary between things um whereas the way that we apply this is still in terms of statistics and Dynamics and processes so we don't really deal with that aspect here should we really expect things this is my point here this is the question that I'm asking should we really expect things in active inference which is a process Theory I mean I understand that the spatial boundaries they are a matter of convenience but can we really expect them to sort of you know divide what what really is like the central nervous system when does it really transfer into B into uh uh into the actual uh flesh of the body and these spatial transitions they sort of happen in degree and if we look at the borderline we will we will be looking at that forever and ever and we look at neurons and then atoms of neurons and so on and so on we will never be able to really find the delineation this is where I think that the idea of temporal Markov blankets sort of is a good idea just to briefly sum this up right so Markov blankets is of this particular node um is constituted by its parents and its kids and the kids of the parents of the kids here so anything that is around that will not hold any valuable information about this particular node again I'm probably preaching to the choir here um but here I'm basically asking so the delineation is smart because it's sort of you know it um it gives us tractability right it makes it tractable we can start thinking about the Dynamics between things um but in the context of time we might ask can we apply the Markov blankets to the temporal structure because we already drew them as conditional dependencies so we sort of asked in our paper what are the statistical boundaries of the present moment so you I just show you this one here before um of the integrated continuity drawn up here as a network of conditional dependencies so if I were to sort of look at this particular Point uh the the event B well we will find that if we try to map out this Markov blanket we will find that well the information necessary to understand this particular node is actually constituted uh by a bit of the present and sorry a bit of the past in a bit of the future right so basically we're kind of cheeky doing this when I think about it because we're sort of dressing the structure with a blanket that that um that happens in time and I think that obviously the the uh how can I put this this is unconventional and which means that this will have to be um there will have to be some limitations uh compared to the the conventional way you would use Markov blankets in spatial conditions uh special uh relationships um first we find that well all the information needed to understand the

present moment um well we find them to comply completely with uh both principles of of phenomenology we basically get the interpenetration of Passenger and the Pres on the future and we also get this integration between the objective and the subjective time so we get an integration in this direction and we get an integration in this direction here um so that was actually our very first surprise so the second surprise when we did this it sort of occurred when we try to understand to understand um what does it take to update the present moment right and to actually propagate the whole model through time and for that we need to analyze the update function of a standard active inference pmdp so yeah probably I should mention that the conventional pmdps in robotics they don't use the same kind of of update function as we use it in in active inference and because they're much more explicit about the reward functions and so on and active inference uh obviously caches all of that out as a free energy minimization so anyway the Dynamics of the present moment are constituted by two potentials we have them up here and a single retention which is down here and basically you will be able to get this particular equation and once we start to sort of um because these are factors so if you if we if we sort of want to turn that into sums we need to convert them to the logarithms instead and basically if we just keep this one in mind and put it over here we see that the first line is our derived model from the slide before and of course like any anyone who's modeling we have to engage with this very Central exercise which I like to think about as providing quality to quantity and I think this is a very very particular problem that the researchers in the field of computational phenomenology have to deal with as well so keeping this exercise in mind because I will I will have some comments about this towards the end so we have the hidden protection at t plus one and the hidden Primal impression at T and T plus one basically what you see here um and then we have the observed retention at T minus one this is basically what we derived from from our uh model from before now uh the second line here is basically the update function of of the active influence from DP and once we actually just strip it away if uh strip it from all its uh action policies you noticed here there are no action policies in this update uh I will return to why we did that then basically what you have is the well it's pretty straightforward actually you have the prior which is the forward message passing and you have the predictive which is the backward one and you have the likely outcome our notation here with the double timeline the Double Dimension of time in the single notation sort of allows us to understand better what is actually going on here if we map um the phonological version or equation that we derived onto the existing update function and so I've omitted here a few equations that if you're interested you can find them all in the paper

but essentially once it's all mapped out it appears as if the update function can be translated into kind of a prior protection a predictive Primal impression and a likely retention so very similar to how protection was brought backward the retention here is brought forward in time linked with the likelihood and the and the hidden state um the discovery or a surprise as you rather say is both that the model appealed to the same temporal structure as the the active inference model although in a different way so I just took this quote from uh from our paper because this is how we sort of reason about it but this gives rise to an integrated structure of time where the past depends on prior predictions and the future of of my likely retention which speaks to different logical understanding of experience so just as in the hostility analysis the retention Primal impression and protection jointly make up the living present then we refer to uh the set of inferences presented here as the living inferences which is sort of you know a work like here between formalities so since they are drawn this they serve together uh sorry since jointly they serve to describe within an active inference model the phenomenological process that gave rise to the living present so this is basically uh me working my way here to to the end um we don't have the answer here um I still think that this is very clear here but we have a suggestion for an answer which is the continuity that we experience is due to the interpenetration and synthesis of an objective and subjective temporal dimension so yes um can we extract principles that was basically our answer here how does the continue flow in Human Experience emerge well through the penetration and synthesis um of two temporal Dimensions can we extract principles from her shoulder and temporality yes we can easily extract principles from houstonology we just need to note on the what conditions this is done and I think this is an important note because you remember when I said that there's this quality to quantity exercise that we all have to engage in so basically my view is that computational phenomenology if it wants to contribute to science this would be an extremely important exercise but when and how quality is attributed to a specific quantity it needs to be extensively and like very very clearly uh like elaborated argumentation because um so you can ask why does this particular parameter here right um like typically in active interest that would be the temperature you say that oh this this particular parameters it represents attention well does it behave like attention and to what extent because it will have a limit and we just need to know when is that limit um basically in our case we saw that the temporality can be modeled but we need to know under what circumstances so in our case we assume that the present moment is not a static Frozen moment in time just like space but it endures there's a duration right and it's essentially this Insight that allows us to create such an integration between the two temporal

dimensions so finally I'd like to think that we demonstrated the potential of active inference in being used as a as a Point of Departure to model basically the flow of subjective experience um but I think that we have yet to begin analyzing anything beyond that um we basically need to start thinking about including action policies so that's motion and action and of course essentially we would arrive at something that looks like content right um so so I guess this is in a way our rigorous approach and and humble uh uh start with with will the very Basics um and I think that this very slow approach is is exactly how it should be because we should be cautious um about this relationship between phenomenology and computational modeling just a few words here about our future work and what we were working on now and this is um actually um this is an example of of um one of the things that we are working on I will say that right now we are focusing on another um uh perspective of our work which has to do with the direct integration of motor intentionality and an action but basically in this particular project we're sort of trying to map out the exact um let's say structure of of time in any computational model right because when you analyze the update function of any computational model it will reveal its temporal structure like how does it bring the past into the future right into this propagation time um so you could ask like what about the CNN's right uh convolutional new networks what about recurrent neural networks or Auto encoders or all the other kinds of algorithms how do they function like how do they update their function and I think that um by analyzing all of these different algorithms and their update function we could sort of map out all of these temporal structures and understand better how they are propagating in in in in space Through Time so um this would be I think my uh my last slide here yes thank you for your attention that was much longer than I am sorry about that was I muted the whole time longer in time or what exactly do you mean by that well I would say subjective the time was very short uh objective time was much better thank you both for this really excellent excellent dyad of presentations and I will uh pass first to Dean or Ali whoever would like to just jump in with a first thought or reflection please feel free go ahead Ali uh well thank you so much for your amazing presentation uh gosh I have so many comments and questions to ask um I don't know where to begin but you see because I'm also deeply interested in um process philosophy as well and if not necessarily a whiteheadian process philosophy but generally thinking in terms of flows and flux and so on as opposed to substance and stasis uh but uh in my view uh well historian phenomenology uh first the these conceptions the flows and the temporality and retention protection and so on uh is a step toward uh kind of process thinking right but ultimately uh I believe that certainly on

philosophy or more precisely historical phenomenology uh fails to escape the uh the Trap of substance thinking uh because uh you see uh if I read a direct quote from Husserl himself he says that time is like a flow but with divisions in other words not like a flow at all and in fact the opposite of the flow so uh the reason he's he doesn't bestow uh the kind of this kind of Primacy to the temporality or continuity is because obviously it results in a contradiction in other words uh if I mean as if what constitutes objects um uh if time is what constitutes objects in time obviously that cannot itself be an object in time so uh he needs a more primary uh way to or more primary concept to account for uh that kind of division in temporality uh namely the absolute flow uh but uh well there are some arguments regarding uh whether we can escape this uh apparent or real uh contradiction or not uh which I'm not going into detail but at one point I wanted to make is that if we uh also try to bestow this kind of Primacy to the absolute flow or who's certainly an absolute flow don't you think we basically contradict our initial aim to escape that kind of um I mean substance thinking given this uh contradiction that arises that's one of my question uh the other ones um is about um the conception of objective time and subjective time uh so uh interestingly well who saw you you also began your discussion of Husserl's phenomenology with um musical crystals famous musical examples and it's very interesting that probably the most influential music theory of the past hundred years uh namely the Schenkerian theory uh it is fundamentally a phenomenological Theory because uh he tries to theorize how we perceive each moment in music as a culmination of every bits of Music we've heard so far until that point and on the other hand the anticipation of what is to come and the others so each point of music is a kind of combination of all the elements of retention um Primal Primal Primal moment and also the protension but there are some criticisms put forward especially uh by Gerald Levinson regarding the feasibility of this kind of description uh because of our limited capacity of our short-term and working memory in trying to at least consciously grasp all of the details of the past and also our limited capacity to predict every um a possible future or every every anticipation from that single point uh so my question is uh and also uh there is another criticism regarding the uh the ignoring the uh per elasticity of perceived time in Schenkerian uh Theory and also in his early and phenomenology so my question is how would you account for this kind of apparent elasticity that all of us experience in um the subjective or manifest time and my very last question sorry it took me so long and my very last question about the is about the objective time which is do you have any particular justification or any particular reason for choosing one among many of the physical theories of objective time so for instance we have uh view from quantum gravity review from relativity

which basically are seeing can be seen as kind of contradictory because in general relativity uh we we allow for the time for temporality to be fundamental but in quantum gravity temporality is a kind of emergent property that emerges from uh to each wheeler equation so uh is there a specific reason that you you chose you decided to choose Carlo ravelli's uh I mean conception of objective time uh again apologies for my long-winded questions I'm looking forward probably a little bit better so we can see each other when you start the when I think some of the first questions were about the about the phonology or I can actually I have a few comments to that and I think that if I may uh um yeah go ahead go ahead I'll talk some comments but go ahead yeah so I would like to just start by making this the the distinction between um time perception and the temporality and phenomenology um because um temporality is more basic than um time perception time perception is actually a skill it's actually uh something like it's it's sort of like attention and memory and so on it's a skill set we have and it's something that we are performing uh some are good at it uh some are very bad like me um but in any case temporality in phenomenology it talks not about how you um your skills about time perception it's more about the actual unfolding of anything like any experience whatsoever like me now shaping these thoughts to sort of convey them I'm not thinking about time perception I'm actually it's happening and there's like this unfolding of time constantly this continuous flow right uh I have whenever I talk about it I like to think about Lee smallen's reality of time uh I think that there is some truth to to to his way of reasoning about time the fact that it's real and it's actually more real than space and space is actually the emergent property of time that's actually by the way from from uh category Theory and geometry Theory as well um that's where where that comes from but because if you think about it like that right if you think about the time being the thing that sort of pushes us forward right there's like this action this I would I wouldn't even say this in involuntary uh action like you're constantly doing this it's constantly happening so for me I would make a clear distinction between time perception and temporality now if I just jump directly to the end of the question because um I I love to deal with with the you ask why did we go with color rovery's thermodynamic Arrow of time um there's a good reason for that because in the beginning um I believe we discussed briefly the uh whether the divided equations because in that equation you completely take away time right you just say well basically you can explain any kind of Dynamics and and change without any variable of time you just need special uh parameters um I personally think that that's uh great that shows that math Works uh doesn't mean that that's how it works um to me the way that I think about it is is more about um so uh it doesn't matter what I think about why

do we go with the thermodynamic uh color of English thermodynamic approach because he starts with entropy and that sort of builds in uh perfectly well on on our approach and our analysis of active influence being built on on entropy being however another kind of entropy it still Builds on the very same let's say ontology right it's built on the same idea and it fits that idea um could still be wrong in any case if you go with David and Wheeler's equations you still have a another dimension of time that is different from the subjective one and I think I should be clear here that I'm not saying that um they exist independently I'm just saying that it's two different things and we can think about them individually and there is an advantage in doing something like that because then you can start looking at how they integrate afterwards so this is merely convenience um there is no such thing as only a subjective time without objective time and the other way around this is not part of of the paper it's a Way Beyond the scope of the paper anyway yeah kind of lashing into to Zach's comment towards the end is that when we were writing the paper we're discussing slightly about how we should address objective time and in the end we decided to use catalog about this framework as such as just a matter of convenience I think at least I'm very much instrumentalist in that sense that you can have a lot of different stories of how physical time works and that's important and that's okay but in the end I don't think that affects too much our overall purpose which is to say there's an integration between an objective sense of time take color really take quantum physics Etc and subjectivity which in which we are pretty much committed to the hotel in framework which this brings me to the other two questions concerning the first one about uh kind of the relationship between process intology and reselling phenomenology I think the important thing to note here and this is my interpretation to be honest is that when we talk about Consciousness and time Consciousness and the absolute flow within the consolidian framework it's important to notice that it is not a thing and I think it's also an important thing to keep in mind when we address how are we to explain Consciousness scientifically it's uh I do believe that's a sort of implicit assumption within a lot of the literature especially the earlier one it's assumed that Consciousness insofar is a phenomenon it's also a thing that has to arise from say neural processes or neophysiological process or whatever kind of frame you're subscribing to but it's not really that and that's the nice thing about phenomenal this especially in phenomenology is that when you go to the kind of the deep level of whatever is happening subjectively you won't find a thing you find what he calls the absolute flow which does sound like a thing as a river and he does say that it's it's a flow but it's not there's always this hesitation the literature because he he notice that every inflow what we experience there's

always this uh there's a lot of stuff happening at the same time and in time but it kind of keeps a structure which is retention Primal impression and protection so there's this this kind of paradoxical nature to the absolute flow it's Pure Flow but it has a specific structure but when you look at how that structure functions and that's kind of the the step that he takes from what the so-called static to the genetic or dynamical phenomenology is that this structure is just I said processes that are always happening not things it's it's not that retention is a thing but in notes literally process is happening synthesis as he calls it and that's what we think it's it's interesting to connect to the active inference framework or any other modeling framework is that then you don't really have to capture a thing say equality assay uh the redness of the red but actually you're just trying to capture processes with processes so you have retention and you have inferences that's it you don't have real things you have to stop that is happening through time and you don't really have to commit to any sort of uh object ontology you have just processes moving on through which something more stable arises but it's not a thing once again when we experience say the bottle it's it's a thing but it's always given itself in time it's always changing I can see its movement I can move around all this time that's the nice thing about it and concerning the other question uh which was the one and uh the limitations on uh okay working memory about the elasticity of time perception can you elaborate more about the the concept of elasticity as you were using it sure uh you see uh The Familiar phenomena that we experience when we somehow experience this speeding up or slowing down uh the experience uh or living present that's basically what I mean by elasticity of time perception I I often wondered because certainly when we talk about subjective temporality in the hotel incense that's not really what he's trying to capture that's quite clear he's simply when he talks about the structure of time conscience he is very exploited we're talking about formal structure that remains ugly invariant through all experiences but it's true that there's a sort of flexibility to how to experience the the world and this is literally a speculation for me I have to sit down and think more about it but I think there's an important connection between up activity and temporality here so there's there's uh kind of uh especially in activism there's this discussions about how effectivity and time Consciousness in the Australian or phenomenal sense linked to one another for example this is not an activism Explorer Matthew Radcliffe talks about styles of anticipation we may anticipate something with anxiety or or with hope or with fear and that probably stretches out not only that we anticipate but how we anticipate and that's that's very important it seems to be how anticipate something and what we're anticipating something and it this also kind of constitutes our general um openness to the

world because say I'm sitting down completely bored waiting for something not only the the experience of time kind of straight Islam sources out but also you know there's a special kind of effective atmosphere to our world but once again this this kind of opens up another question about for computational neuropsychology which is okay now we have inactive inference this kind of clear-cut temporal a structure that links up to cultural and phenomenology how can we introduce effectivity into the picture and there's of course a lot of work and effectivity emotions at it more generally in the active influence framework but that's an interesting question how can how does this relate to our temporal model I don't know if Sac has anything else to add I was just uh reminded of a few things there uh I think you're you're totally on point with with everything you said I just want to emphasize the fact that the stretch and the plasticity of time is still in my opinion something of a skill that we have right it's an impression and that impression is something like that we that we get at um and just because I have that experience doesn't mean uh if I had like a twin and we were doing this not even just a friend and we were doing the exact same thing that stretch will be very individual and it that just means that it depends on well on who you are and your skill at doing that I I cannot really differentiate between time perception and the plasticity that sort of is implicit in time perception I cannot distinguish that from like other skills like being able to remember something or directing your attention at something um because that's exactly the same thing we don't have the same skill like some people are very bad at suppressing a peripheral information others are not and it just continues like that um but but I I do have a um a point um so do you guys know James Tyrell the Light Designer uh artist American Artist he's a fantastic Light Designer I highly recommend you go see all of his work um basically he has he studied the eye and vision so never a vision he's not a vision neuroscientist but he studied vision for many many years and he's a Light Designer and basically he creates these huge installations everywhere um and one that really caught my attention was in Japan uh I think now it's seven years ago where we went into this pitch dark room right it's a big Temple that was uh reconstructed pitch dark couldn't see anything and you just had to touch the wall and you had to follow the wall um and then suddenly you feel like okay now there are other people also holding the wall and you can you sort of feel other people you go okay I need to stop I don't I shouldn't follow the wall anymore and at some point you're just staring out in complete darkness the space suddenly emerges your eyes adjust to the very very low uh light uh well the locks I don't know the exactly illuminance I suppose in English um the very low illuminance you start to adjust and this can only happen over time I think that this this sort of is an example

of temporality sensory impression content uh I wouldn't say call there because I don't buy into that um I'm totally uh I'm more of a Time person I don't know if you get the impression um but I have a question for you as well Ellie if I may because this this example that I just gave here with with the James Terrell and I'm an architect myself and usually I give that example with the spaces did you would you say that it's the same experience or would you say that those are two very different experiences uh well certainly there are similarities between those two experiences I'm not sure if they're identical or even they can be uh discussed in uh I don't know in the same terms but um going back to your um point on the temporality and also this skill the the discrepancy between the diplomacy of the scale the Primacy of the temporality as versus the scale of time perception uh well I certainly agree with you on that uh but I also believe that you see uh I don't think at least in my in my opinion that the temporality or the absolute flow or the at least you're certainly an absolute flow is the most fundamental a fundamental concept that needs to be grounded because uh there needs to be uh even more fundamental prior to even existence of that temporality or that flow right uh which I'm not going into details about how that fundamentality uh can be accounted for or it can even be discussed in terms of our observed phenomena or our experience phenomena but I wanted to thank both of you for your extensive response you provided and helpful responses I wanted to ask for your permission to uh I I wanted to discuss these matters a bit further with you so if you permit I wanted to email you about them and discuss them a bit further because that's something that I'm deeply interested in so again thank you so much because Ali just mentioned this that he doesn't think that the world is kind of the basic thing and I I don't know to what extent I agree but something I do think that happens is that at least in the Australian framework there's this kind of conception that Consciousness or subjectivity more generally is this kind of layered or hierarchical structure of which you have to find literally the bases and go from the ground up so to speak and I think there's something true to that but it's not completely it's not as hierarchical as we would put it because once again this brings us back to the concept this relationship between up activity and temporality for me I would say app activity inform centrality and temporality in forms of activity are co-emergent but how can you affect it because you have a body and then you introduce embodiment into the picture and you've introduced environment to the picture you open up a lot of like it's you open up the Pandora's Box almost so maybe this whole idea of the most basic stuff it's a bit of a red herring but I also think there's something valuable and fun would agree on this bit is that temporality is it's very abstract it's just a formal structure which underpins all experience regardless of

whether it's the most basic structure or not that's a different question and that's all I want to say for now awesome all we'll pause in time and move on Dean please where does this take you well I'm like Ali a whole bunch of places I'm not like Ali in the sense that I do place a very high priority and emphasis on timing and perspectives um that doesn't mean that Ali and I don't get along fantastically we do it's just that it's that different perspective thing that I think gives this whole topic of certain richness I want to touch for a second on on Zach's uh teasing out the word skills because I think that's really really important here I'm my background is sort of in the Practical aspects of learning um so I I tend to zoom out on not just the particulars but the whole picture I want to just say before I ask my question is I thought it was a fantastic compare comparative analysis in your paper and I thought your your presentation and your sharings today were also very clear in terms of being able to show difference insane which I think only comes about when we account for the temporality not that I'm not saying measure it but at least have a certain appreciation of it so here's my here's my thing practicalities in other words the actions and active inference are not necessarily always a capture that that could be one form of an action but it's not the only one there's there are plays there are creations and compositions which we looking at category Theory are quite intimately familiar with now one of the things that I tried to bring to the conversation about a decade ago I was working with a number of people in complexity science and complexity research and they were a huge match Arana and Varela fans and for about two years all I did was have conversations like the one we're having today with complexity scientists and one of the things I tried to bring to them in terms of understanding the possibility of applying skills as learning were number one you first have to be able to get people to pay some sort of attention have an awareness of of what they are embedded in what their Niche is what is being signaled to them Etc number two you have to be able to link and combine and I know that in a lot of formal learning settings that is that that Association piece that skill of being able to do that sort of compare and is this same Is this different again category Theory stuff there's a heavy emphasis placed on that then there's another skill which is to be able to set a rule to be able to scale up around what is consistent what is seeming to be able to have a certain shelf life and things that are changing rapidly because we you guys talked a lot about change today so that's basically an understanding you can you can know in English where that came for me basically standing under the abstracted state the heuristics the little sort of rules of thumb that you apply as a skill subjectively and then finally and this is one that's left off of most out of most conversations there's a particular type of software and all of these are self organizations even if you're being prompted by the

teacher you still have to take up the skill of organizing yourself but there's a fourth type of skill which is a particular type of self-organization which is being the capacity to transfer recapitulate in a timely manner and I think that's where your paper is opening up a bit of a Pandora's Box it's not just a comparison of objective and subjective time it's how how do we know when to reapply the rule and I think from a learning standpoint that's a that's a big moment it's not just a Eureka moment but it's a sense of do my inferences now withstand the scrutiny of the environment the environment generally not just the niche so is the niche capturing us or are we still free to think about how that next move fits with all the prior moves and I think that requires the representation that you guys had in your paper that essential diagonal line and so I think I think that's what you did I think what you've touched on it and put into into defensible form is okay so there's there's all these skills and do we cut ourselves off at the at the at the more phenomenological ones are you paying attention are you making associations or are you freeing yourself up to be able to have these rules but then know when to re-purpose them as the context changes so I want to applaud you because I I've had many many conversations for over a decade where there's a whole bunch of people who do not want to talk about the third and the fourth skill because it's hard to measure but I think you guys have brought some some ability to compose that and so that's all I wanted to say I don't have any questions just want to give you a two claps thank you thank you very much that's that's uh wow thanks thanks a lot the I don't even know what to say because uh I guess to Juan Diego and I it was actually I think over a Twitter conversation we sort of realized a few things and basically well linked up and started working on this and apparently we have a lot in common in terms of philosophy uh well and also other things but um interestingly we found out that there was actually a you know plausible retractable uh relationship that we could exploit so well uh thanks thanks a lot I'm very happy to see that there's someone else who is also like a very temporal person right like think thinking about things in time and and events rather than things and I think this is a very important distinction yeah for sure thank you for that coming Dean and I also think that it can kind of speaks uh to the whole temporal thing we're talking about precisely because what we learned we learn through time and that's kind of obvious but but you have to kind of address how The centrality Works in order to eventually talk about learning and pulling rules and being understanding our environments you were saying kind of understanding how to apply these rules like how to apply different skills in a proper environment adaptively so to speak and something that first uh kind of attracted me to the active inference and fep context possibly these possibility of making sense of how through

updates to genetic models um you can try to make sense formally of the of these skills and this process that underline what it's like to be you know an organism and a human being you know changing world and once again it brings us back to change temporality all this stuff it's all connected I just can I just do one quick follow-up Daniel because I know I want to make sure you have some time as well I think what people struggle with is they can name it and call it transference and oftentimes that gets conflated with understanding and wisdom and that isn't necessarily so that timely way of being able to recapitulate something is an incredibly important skill if we get that wrong we oftentimes you know materially fall off a cliff but whether or not we can see it as both discretized as its own entity its own particulars and part of this massive whole that requires all of these priors as well being able to juggle that I think is again that's what you've opened up here and hopefully people are able to then take that information and test it because that's the best thing to do I in fact this morning in a in another separate conversation altogether um I was asked how to take some of the practicalities and the skills that you guys talked about in your paper and apply them in real real time and it's not really really hard if you have at least those four skills it is very difficult if you stop at the associative level or if you if you stop at the I built a rule now and I know how far that rule will extend so that's my my final say for today go ahead sorry Daniel oh it's all it's all great um so many threads weaving together so again to to both presenters and authors it was it was excellent the the dominant uh sense or impression is is that it's very evocative what you've presented the idea of a generative passage not just as a touch upon generative models generative AI but also this idea that our travels are generative like whenever we see a friend and we take the train we always have some story to share and like something that happened along the way I know that's not necessarily how it's used but it's just what it evoked when you describe formal dynamical models as generative passages it's not 95 Theses to pin onto the door it's something that evokes and you even spoke about taking different perspectives on your work rather than your work advancing a perspective and then to hear about the connections with the um light and sound art and visual art and uh the the notion of architecture and your background Zach and and the interplay between space and time time and the way that sometimes we can look at a blueprint and see it as static or even the constructed embodied building as a static object when it also moves through time um then one more impression and then a question finally the um the vocabulary and the the delicacy of the articulation with the protention retention and the Primal impression or the living presence and then the way in which graphically you arranged it um along orthogonal axes like rows and columns that

allowed us to slide through on the diagonal um it gives a lot to work with and my question is where is the action perception Loop or how do you believe that this time Consciousness and Associated methods and skills how can we then go back to potentially earlier or contemporary representations of the particular partition under the fep or just the active uh inference action perception Loop how can we see that or be with that differently with the advances that are happening literally in the last year in this area oh man that's right on I mean spot on this this is exactly the kind of research I do basically um I have like I guess two paths in my research um I think eighty percent of my time is Empirical research doing Neuroscience uh and basically doing experiments all the time and one way that I translate all of this all of these insights that I get from from working with someone like Juan Diego is to try to understand them as uh what I what I like to think about as a sensory motor entropy uh basically talking about it as the interaction and the loop that naturally occurs not only in actual perception but also in past and present and if I may just briefly um share my screen again I would like to show you that um this last aspect I think I hope you can see my arrow I'm not sure um I'm just gonna take the laser pointer just to be sure that this last Point here I think this sort of maybe over here is better you can see that the retention and the pro tension sort of uh correlating or sort of you know um meeting here in the middle and sort of creating a loop through time so if we have to jump from this point this is the update step some from updates that one to two you sort of need to create this interwoven link between them this is naturally creating a loop between uh well action and perception because actions breaks you what brings you forward in time perception was part of the retention and basically there will be this naturally occurring Loop just from basic principles of temporality it's like you cannot get around that action perception sort of emerges from from temporality all naturally um so in my opinion the way that I think about it is as sensory motor entropy and basically you can look at that and the amazing results that people are finding now around the thalamus which is basically the essential organ of the brain the subcortical structure actually functions as a bridge between major brain regions right so for you to actually perform any kind of action it needs to use the thalamus as a bridge which means that it's intrinsically related to the way things change and there's like this I guess you can think about it not as things but as events right so to give an example like Juan Diego did with his uh this is my cup and the fact that my car continuously exists is not the thing that exists uh all the time but there are events occurring in which this one continues to exist right so it's more about thinking of the evolution of time being highly correlated with itself rather than thinking about things existing in time but this is like a reverse way of thinking about it

and the typical strategy of of active inferences clipping things on the head and see if it actually still makes sense in this case I I genuinely think it does I don't know if I if I answered your question yes thank you Dean yeah I would I would hold just to be able to hold up two things again I would say that there is the bridging piece and there is also the high occupancy vehicle Lane as opposed to the intersection like a lot of the ways that we represent this stuff is when we get those X and Y um vectors Crossing one another but there's also uh let's get all four skills into one onto one platform and remove a lot of the impediments and that's that flipping part I I would agree exactly with what you're saying you have to have both one as a metaphor as a metaphor it's all that's all it is metaphorical but if you can get all four skills in one place and then down down the distance you'll shorten the amount of time it takes to arrive at wherever you're going thank you one um and please feel free to give kind of your your your closing thoughts and next angles as we come near I don't even know whether to call it an ends so I just wanted to to to Daniel's question because I thought was a great question one I didn't think about at least in the specific terms with Zach we've discussing precisely kind of the the where action fits in all the story and this of course brings us to the action perception Loop and I guess I'm going to do what we usually do in our collaborations with Sac which is I'll bring the phenomenology and see whether it fits and and I think it fits quite well because what we did in our model is to show that there's precisely this boundary portal boundary to the to the updating scheme of active inference which integrates pattern future but we and we kind of relate this to protection rate tension Primal attraction and you have this nice living present what is fascinating is that when you go a step further away from the pure temporal form of experience you have an embodied being whose anticipations whose protections are not some not simply the tones are coming from the melody but what do you do with them or what do you do so if I see my bottle again this bottle appears as something I can grasp and you can think about this in terms of importances action possibilities it's in that very clear in the phenomenological literature and also I think in the active influence literature when we talk about action policies and you know specific importances all this stuff and it is fascinating because then you get action just by introducing the temporal aspect of of active inference but this action is always informed by prior observation so by literally perception and that happens well in phenomenology all our actions are important but what we are seeing what we're perceiving more generally and what what we've perceived in the past and you can just kind of unfold that Loop and get a very nice correlation between action and perception through time so the actual perception look is not like this monolithic Circle it's something that it's more like

a spiral so to speak and I think our model speaks to that anything to this it really would make uh worlds of difference to call it an action perception spiral if it's just gonna be a visual metaphor and rely on our cultural understandings of shapes why not firstly hold up multiple and then secondly bring in um this kind of perhaps even double helical or triple helical nature of of unfolding and three makes a party it's awesome Ollie and Dean first any closing um thoughts and then the the authors with the final words I'll be really quick I really want to thank you both for uh sharing the paper and uh the great presentation thank you Dean Ali yeah as I said it was a really fascinating paper for me personally because uh I was really obsessed with um process thinking processual and all kinds of dynamic thinking so again I really wanted to thank you for writing such an amazing paper uh yeah thank you so much awesome and so to the authors any closing thoughts or where you'll be heading now or what you're excited what you're pretending I guess first of all I would just like to thank you because it's been an amazing experience an experience simple experience actually just to discuss our paper with you thank you for reading the paper thank you for enjoying the paper and happiness it's it's been lovely and uh I I think it's quite clear that the way to go now is to start talking about action and see how it fits and hopefully sooner right until later you'll see something about this because it's probably what we're discussing with Zach more recently yeah I also would like to extend this thank you uh it's fantastic to just I mean the whole active inference Institute is just in my opinion the right thing uh it's really amazing to to follow um so I'm very happy to to be part of um Beyond uh well besides the the active aspect of uh hopefully the next paper uh I'm also looking into phases and looking at active inference as if it did not work with States because I think that's too spatial and it's making things static but actually moving that into phases into a continuous space um is something that I'm trying to establish now uh trying to start up a collaboration with Ryan Smith about that but it's still a really early work in progress so hopefully we'll see something from that project in a few months time hopefully but but thanks again thank you awesome thank you all for joining thanks everybody for listening and you're always welcome back so till the next bye thank you